# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

## ANSWER KEY & MARKING SCHEME SET 04

**Full Syllabus (Chapters 1–13)** | Class 8 Science | Maximum Marks: 80

### SECTION — A (Objective Type) — Answers — 20 × 1 = 20

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(d)** way the dough enters the oil | A controllable factor |
| 2 | **(d)** something we measure | Observed or measured outcome |
| 3 | **(b)** Ananda Mohan Chakrabarty | Cleaning oil spills |
| 4 | **(d)** Diabetes | Lifestyle related |
| 5 | **(a)** symptom | Felt by the patient |
| 6 | **(c)** heating effect of current | Filament glows |
| 7 | **(d)** positive terminal | Positive electrode |
| 8 | **(b)** irregularities of surfaces in contact | Even smooth surfaces have tiny bumps |
| 9 | **(c)** heats faster than water | Creates low pressure |
| 10 | **(c)** greater | Larger difference drives faster air |
| 11 | **(d)** liquid and gas | Particles move past each other |
| 12 | **(c)** new substance with different properties | Compound differs from components |
| 13 | **(b)** a metal with other metals or non-metals | Solid mixtures |
| 14 | **(c)** 1 g | Density 1 g/cm³ |
| 15 | **(b)** also reflect light (diffuse reflection) | Light scatters in many directions |
| 16 | **(d)** tropical year | Basis of Gregorian calendar |
| 17 | **(b)** organic manure | Vrikshayurveda advocates it |
| 18 | **(d)** eagle | Fourth trophic level |
| 19 | **(b)** air | Mixture of gases |
| 20 | **(d)** fertilisation | Forms zygote |

### SECTION — B (Very Short Answer) — Answers — 8 × 2 = 16

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 21 | Cause diseases in humans, plants and animals (1). Spoil food (1). | 1+1 |
| 22 | Diabetes (1). Cancer or asthma (1). | 1+1 |
| 23 | Current flows through it only one way (1). Long lead must go to positive terminal (1). | 1+1 |
| 24 | Fast air lowers pressure between them (1). Higher outside pressure pushes them together (1). | 1+1 |
| 25 | Elements with properties between metals and non-metals (1). Silicon and boron (1). | 1+1 |
| 26 | Ice layer insulates the water below (1). Fish survive in cold weather (1). | 1+1 |
| 27 | They slow winds and waves in storms (1). They are home to many species (1). | 1+1 |
| 28 | Asexual: single parent, identical offspring (1). Sexual: two parents, variation in offspring (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 8 × 3 = 24

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 29 | Fair: only oil temperature is changed (1½). Unfair: temperature and thickness both changed so cause is unclear (1½). | 1½+1½ |
| 30 | Push an empty bottle into water (1). Feel an upward push (1). It bounces back to the surface due to upthrust (1). | 1+1+1 |
| 31 | Solids: iron, ice, wood (1). Liquids: water, milk, oil (1). Gases: air, oxygen, methane (1). | 1+1+1 |
| 32 | Plane: same-size erect face (1). Convex: smaller face (1). Concave: enlarged face (1). | 1+1+1 |
| 33 | Stars in the background of the Sun marked the seasons (1½). This helped create accurate calendars (1½). | 1½+1½ |
| 34 | Reduce waste (1½). Save resources and energy (1½). | 1½+1½ |
| 35 | Do not touch bare wires (1). Do not use mains supply for experiments (1). Disconnect circuit when not in use (1). | 1+1+1 |
| | **OR:** It still contains acids and metals (1). They can cause fires (1). They harm the environment, but can be recycled (1). | 1+1+1 |

| | | |
|---|---|---|
| **36** | Bandipur – Karnataka (1). Manas – Assam (1). Chilika – Odisha (1). | 1+1+1 |
| | **OR:** Dead matter would pile up (1). Nutrients not recycled (1). Plants and ecosystem would suffer (1). | 1+1+1 |

## SECTION — D (Long Answer) — Answers

**4 × 5 = 20**

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| **37** | Ayurveda: balance of body, mind and surroundings (1½). Dinacharya and ritucharya routines (1). Variolation against smallpox (1). Modern vaccines are safer and tested (1½). | 1½+1+1+1½ |
| | **OR:** Typhoid by contaminated food and water, so boil water and keep sanitation (2). Dengue by mosquitoes, so avoid stagnant water and use nets (2). Community drives and awareness (1). | 2+2+1 |
| **38** | Warm ocean water heats air (1). Rising moist air condenses (1). Heat is released lowering pressure (1). Earth's rotation causes spin (1). Eye at centre, strong winds around (1). | 1+1+1+1+1 |
| | **OR:** Thunderstorm over land (1). Cyclone over warm ocean (1). Cyclone has eye (1). Cyclone has stronger winds and storm surge (1). Thunderstorm has lightning and thunder (1). | 1+1+1+1+1 |
| **39** | Gas dissolved under pressure (1). Opening releases pressure (1). Heat reduces solubility (1). Gas escapes as bubbles (1). Drink goes flat (1). | 1+1+1+1+1 |
| | **OR:** Density less than water floats (1). Greater sinks (1). Wood example (1). Iron example (1). Ice example (1). | 1+1+1+1+1 |
| **40** | Terrestrial examples (1). Aquatic examples (1). Abiotic differences (1). Organisms (1). Dependence between them (1). | 1+1+1+1+1 |
| | **OR:** Water decreases (1). Plants dry (1). Herbivores starve (1). Carnivores decline (1). Decomposers also affected (1). | 1+1+1+1+1 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTER